Element K: Reflection on the Design Project

Project Title: MJ’s LAST Stand (Echo Rack)

#1

Reflecting on the Project Learning

What did you learn from doing this project? Think about what new things you found out while working on it. For example, "What new things did you learn about building things or solving problems during this project?"

I learned a lot from this project. For example, I learned about pricing and budgeting. When working on our project, I tried my hardest to keep the cost relatively low. Of course we all went into it willing to spend at least $20 each, but we weren’t sure what everything would really add up to. Eventually when I bought all the materials, I think the total was close to around $135. I also learned that a lot of things just aren’t perfect, and sometimes you just have to work around the materials that you have or that you could get. For example, in an ideal world, the 1 inch by 1 inch wooden poles would have actually been the right size, but really they were a little shorter than that. There were also some things that I wanted that just don't exist, like a 3 way hinge. (However, that problem was easily solved later.) I also learned a lot about working as a team and time management. For the most part, we worked really well together and got work done, however, there were some times where just a little bit of extra work needed to be done, and I came in during my lunches to help out.

Assessing the Process

What did you learn about your process for solving problems in general? Think about how you did things and what you could do differently next time. For example, "What things did you do well while working on this project? What things could you do differently next time?"

I learned that a lot of our process for solving problems in general is just creating a quick solution that will probably work, instead of taking the time to think about if it will really function the way its supposed to. Fortunately, this has worked out in our favor and we haven’t had any major issues about solving problems. Next time, I think we should take more time to think about any unexpected problems we might face, or take some more time reviewing and evaluating our design before we actually start building it. In part, this was my fault because it was my design that we were using.

Evaluating Expert Reviews and Surprises

What did experts say about your project? Were there any things that surprised you while working on it? For example, "Did anyone else, like a teacher or

scientist, look at your project? What did they say about it? Were there any parts of the project that surprised you?"
Due to the fact that as I am writing this (4/30/25) we have not shown Mr. Waller our project yet, we can't get any real feedback from him. Our mentor Mr. Sczcpanski, hasn't seen our design since the initial 3D model, because for the most part I only really have miscellaneous questions about how to approach things, rather than how I should go about the actual design itself. There have been (I think) only 3 teachers that have seen our design, or at least variations of our designs. The first one is (you'll never guess) Mr. Paz. Mr. Paz was always very supportive of us and helped us whenever we had questions or concerns, among many other things. Mr. Nusum also has seen our project, mainly because we use his side door to stain it outside. He said that it looks good, and that's about it. Mr. Waller has also seen our initial design, and he pretty much told us that he trusts our judgement and that he trusts us to make something he's happy with. The biggest surprise of our project had to have been the warped piece of wood. It was just so unexpected, and we weren't really sure how to deal with it either. However, due to some excellent redneck engineering, we were able to straighten it back out into an acceptable amount to the point where we can just tell people that it's a design feature, not a flaw.

#2

Reflecting on the Project Learning

What did you learn from doing this project? Think about what new things you found out while working on it. For example, "What new things did you learn about building things or solving problems during this project?"
I would say that this project was for sure a new experience to me. Of course, team projects, where we work on making some solution using the engineering design process such as this one, aren't new to me, but this project is basically all woodworking. I have never performed woodworking so I had some things to learn, such as making pilot holes before drilling in screws, sanding and staining, or connecting pieces at awkward angles. Besides that, I would say I also learned when to adapt or improvise when something isn't exactly going according to plan.

Assessing the Process

What did you learn about your process for solving problems in general? Think about how you did things and what you could do differently next time. For example, "What things did you do well while working on this project? What things could you do differently next time?"
Personally, my strong suit was using a drill and using screws. Out of the four of us, (not trying to be bold or mean) I was the relatively most experienced person and gave tips to my team members throughout the build process. Something I could definitely improve on was my staining. I wasn't exactly sure what I was doing and honestly did a

pretty poor job in my eyes. But now I have a better idea and understand that I need to leave stain on the wood for longer than 10 minutes, and also mix the stain before use. Also I would want to use wood glue next time.

Evaluating Expert Reviews and Surprises

What did experts say about your project? Were there any things that surprised you while working on it? For example, "Did anyone else, like a teacher or scientist, look at your project? What did they say about it? Were there any parts of the project that surprised you?"

After completing the gong stand, the reception was quite positive. Our main stakeholder, Mr. Waller, the Band Director, really enjoyed the end product and commented that we should patent the mute mechanism. I am personally surprised the mute mechanism works as well as it does. There are problems with it such as the pedal requires a lot of force to operate, but it is manageable.